

Requirements Specification

Quant Signal Product — Track Record, Licensing, and Platform

Version: 0.2 (Draft) **Supersedes:** v0.1 **Status:** For review **Author:** Amit Kumar Maharaj

Changes from v0.1

v0.1 specified two products in parallel and implied both would be built. That understated the sequencing risk: every commercial path depends on a live track record that takes calendar time to accumulate, and no engineering effort shortens it.

This version restructures the same requirements into **three horizons separated by go/no-go gates**, so that no capital or compliance commitment is made ahead of the evidence that justifies it. Requirement IDs from v0.1 are preserved for traceability.

Change	Rationale
Track record publication promoted from SF-5 to its own horizon (TR-*) and made the first deliverable	It is the long-lead item and the gate on everything else
Every requirement now carries a priority (M/S/C)	Scope must be cuttable when time is the constraint
Acceptance criteria rewritten as binary, testable statements	v0.1 contained criteria that could not be objectively failed
Options 4 and 5 marked conditional , not planned	Prevents premature build against unvalidated demand
Added explicit gate criteria with numeric thresholds	Removes the temptation to advance on optimism
Added effort/cost envelope per horizon	Makes the commitment visible before it is made

Priority key: **M** = Must (horizon fails without it) · **S** = Should (significant value, deferrable) · **C** = Could (defer by default)

1. Strategic Frame

	Horizon 0	Horizon 1	Horizon 2
Name	Track Record & Foundation	Impersonal Research Subscription	Licensing / Platform
Duration	Months 0-6	Months 6-18	Month 18+
Revenue	None	Low, direct	Contract-scale
Regulatory posture	None — nothing is sold	Publisher's exclusion lane	Technology/data provider
External spend	Near zero	Low	Material (counsel, data licences, SOC 2)
Purpose	Generate the evidence	Test willingness to pay	Monetize proven signal

The governing constraint: **a signal cannot be sold before it has been published in advance of its outcome, repeatedly, for long enough to be evaluated.** Horizon 0 exists solely to start that clock and make the resulting record credible.

2. Definitions

- **Signal** — one model output for one instrument at one point in time.
- **Signal Set** — a named, versioned collection of signals with defined universe, horizon, and cadence.
- **Model Version** — an immutable, hash-identified release of model code, features, and training window.
- **PIT (Point-In-Time)** — data as known on a given date, without restatement.
- **Publication** — the irreversible act of committing a signal to the public record before its outcome is known.
- **Gate** — a decision point with pre-agreed numeric criteria; advancing without meeting them is a scope violation, not a judgement call.

3. Assumptions

ID	Assumption	Validated by
A-01	The existing prediction engine can be refactored behind a service boundary without a rewrite	Horizon 0 spike, week 2
A-02	At least 3 years of history can be reconstructed point-in-time from available sources	Horizon 0, TR-3
A-03	Publishing impersonal, non-personalized rankings without charging carries no registration requirement	Counsel, before first publication
A-04	Free-tier or research-licensed market data permits public display of derived rankings (not raw quotes)	Vendor terms review, before TR-5
A-05	Time available is evenings and weekends, not full-time	Standing

A-05 is load-bearing. Every estimate in this document assumes constrained part-time capacity. If that changes, the horizons compress but the calendar-bound track record does not.

4. Horizon 0 — Track Record & Foundation (TR)

Objective: Begin an unfalsifiable public track record within 30 days, and make the underlying signal generation reproducible enough that a future buyer can audit it.

Success looks like: by month 6, six months of daily timestamped predictions exist, each traceable to an exact model version and input snapshot, with predicted-vs-realized computed automatically.

TR-1 — Publication Pipeline · M

As the owner of a future commercial signal product **I need** predictions committed to a public, timestamped, append-only record before outcomes are known **So that** the track record cannot later be accused of hindsight construction.

Acceptance criteria:

- A scheduled job publishes the current Signal Set on every trading day without manual intervention.

- Each published record is immutable; corrections appear as new records with a reason code, never as edits.
- Each record carries `published_at_utc`, `model_version_hash`, `as_of_data_timestamp`, `target_date`, and `universe_id`.
- Publication timestamps are externally verifiable — public repository commit history, or periodic hash anchoring to a third-party service.
- **Test:** an independent party, given only the public record, can determine what was predicted on any past date and cannot detect any post-hoc modification.
- **Test:** publication continues successfully for 20 consecutive trading days with zero manual intervention.

TR-2 — Model Versioning & Reproducibility • M (was SF-2)

Acceptance criteria:

- Model code, feature definitions, hyperparameters, and training window are packaged as one immutable, hash-identified artifact.
- **Test:** given a `model_version_hash` and an `as_of` date, regeneration produces a byte-identical signal set.
- Published model versions are immutable; changes require a new version with a changelog entry stating what changed and why.
- No signal can be published without a resolvable `model_version_hash`.

TR-3 — Point-In-Time Data Store • M (was SF-3)

Acceptance criteria:

- Price, fundamental, and fund data carry both an event date and a knowledge date.
- Universe membership is stored historically, including delisted, merged, and renamed instruments.
- Restatements append; they never overwrite.
- **Test:** an automated validation job fails the build if any signal references data whose knowledge date is later than that signal's `as_of` date. This test runs on every build and is non-bypassable.

TR-4 — Outcome Evaluation • M

Acceptance criteria:

- At each target date, realized outcome is computed and joined to the original prediction automatically.

- Live (out-of-sample, published in advance) and backtested results are stored in separate tables and are never combined in any output.
- Standard metrics computed per horizon: hit rate, mean and median error, information coefficient, and decile spread.
- **Test:** no code path exists that can produce a blended live/backtest performance figure.

TR-5 — Public Track Record Page · S

Acceptance criteria:

- A public page displays live performance to date, clearly dated, with the record's start date prominent.
- Backtested results, if shown at all, appear in a visually separate section labelled as hypothetical.
- The page states methodology at a level sufficient for a reader to understand what is being predicted, without disclosing proprietary feature construction.
- No language on the page recommends any security to any reader, and no personalization exists.
- The page carries no fee, no gate, and no request for payment during Horizon 0.

TR-6 — Signal Engine Service Boundary · S (was SF-1)

Acceptance criteria:

- Model invocation occurs through a single interface with a stable request/response schema.
- Backtest, publication, and any future live mode differ only in the injected data feed — not in model code path.
- **Test:** the same model version, run in backtest mode over a past date, produces identical output to what was published on that date.

TR-7 — Backtest Engine · S (was SF-4)

Acceptance criteria:

- Event-driven execution over PIT data (not vectorized).
- Configurable slippage, commission, and borrow cost, with defaults documented and applied by default rather than opt-in.
- Outputs CAGR, volatility, Sharpe, Sortino, max drawdown, hit rate, turnover, and a capacity estimate.
- Supports the 10 / 30 / 60 / 90-day evaluation horizons as first-class.

- Every run persists its full parameter set and is re-runnable by ID.

TR-8 — Audit Log · C (was SF-6, deferred)

Deferred to Horizon 1. TR-1's immutable publication record covers the Horizon 0 need; a full per-actor audit trail is only required once external parties have access.

GATE 0 → 1

Advance only when **all** are true:

Criterion	Threshold
Continuous live publication	≥ 6 months, ≥ 95% of trading days, no gaps > 3 days
Reproducibility	100% of published signals regenerate byte-identically
PIT validation	Zero lookahead failures in the trailing 90 days
Signal quality	Live information coefficient positive and directionally consistent with backtest on at least 2 of 4 horizons
Counsel confirmation	Written confirmation that the Horizon 1 offering sits within the publisher's exclusion

If signal quality fails: do not advance. Return to research or stop. Advancing a signal that does not work costs far more later than stopping here.

5. Horizon 1 — Impersonal Research Subscription (RS)

Conditional on Gate 0. Objective: establish whether anyone will pay, at minimum cost and minimum regulatory exposure.

RS-1 — Impersonality Constraint · M [POSITIONING]

The single requirement that keeps this horizon in the publisher's-exclusion lane.

Acceptance criteria:

- Every subscriber receives identical content. No output varies by subscriber holdings, risk profile, account size, or stated goals.
- No feature accepts a subscriber's portfolio, positions, or personal circumstances as input.

- No output uses second-person instruction ("you should", "sell your"). Output describes model rankings and model reasoning only.
- Content is published on a fixed, regular schedule regardless of market events.
- **Test:** a code review checklist item blocks merge of any feature that branches output on subscriber attributes.

Note: personalization is the boundary of this horizon. Any request to add it is a Horizon 2 decision requiring counsel, not a feature request.

RS-2 — Subscription & Access · M

Acceptance criteria:

- Subscribers authenticate; content is gated by active subscription status.
- Payment handled by a third-party processor; no card data touches the system.
- Self-service cancellation with access continuing to period end.
- Free tier exposes the track record page (TR-5) and lagged rankings; paid tier exposes current rankings.

RS-3 — Content Delivery · S

Acceptance criteria:

- Rankings published to web and email on the fixed schedule.
- CSV export of current and historical rankings for paid subscribers.
- Archive of all past publications remains accessible and unmodified.

RS-4 — Disclosure · M [POSITIONING]

Acceptance criteria:

- Every page and email carries disclosure that content is impersonal research, not individualized advice, and that past performance does not indicate future results.
- Hypothetical and backtested results carry separate, explicit disclosure wherever they appear.
- Disclosure text is reviewed by counsel before first paid subscriber.

RS-5 — Demand Instrumentation · M

The actual point of this horizon.

Acceptance criteria:

- Tracked and reportable: visitor-to-free conversion, free-to-paid conversion, monthly churn, and reason-for-cancellation where volunteered.
- Subscriber cohort retention measurable at 1, 3, and 6 months.
- A structured record of every inbound enquiry about licensing, API access, or institutional use — these are the Horizon 2 demand signal.

RS-6 — Audit Log · M (was SF-6)

Now required. Append-only log of access, subscription change, and publication events, capturing actor, timestamp, action, and resource. Minimum 7-year retention.

GATE 1 → 2

Advance to licensing or platform work only when:

Criterion	Threshold
Track record length	≥ 18 months continuous live
Live signal quality	Consistent with backtest; no material decay
Paying subscribers	≥ 100, or ≥ 3 institutional inbound enquiries unprompted
Monthly churn	≤ 5%
Counsel	Registration path for the intended Horizon 2 model determined and costed
Capacity	Funding or time available to sustain an enterprise sales cycle of 6–12 months

Horizon 2 is not entered by default at month 18. Failing this gate is a valid outcome; Horizon 1 as a standalone business is a legitimate end state.

6. Horizon 2 — Conditional Commercial Build

Requirements below are **retained from v0.1 and deliberately not elaborated further**. Detailing them now would be speculative. Each is expanded into a full specification only after Gate 1 passes and the direction is chosen.

6.1 Option 4 — Signal Licensing (SL) · *conditional*

Carried forward: SL-1 Signal Set catalog · SL-2 universe definition · SL-3 authentication · SL-4 entitlements · SL-5 core API endpoints · SL-6 bulk delivery · SL-7 rate limiting and metering · SL-8 sandbox and trial · SL-9 documentation and methodology whitepaper · SL-10 SLA and status page · SL-11 contract billing.

Entry preconditions: market data redistribution agreement executed; SOC 2 readiness underway; at least one signed design partner.

6.2 Option 5 — Platform White-Label (WL) · *conditional, deprioritized*

Carried forward: WL-1 tenant isolation · WL-2 provisioning · WL-3 visual white-labeling · WL-4 feature configuration · WL-5 SSO and roles · WL-6 research experience · WL-7 portfolio context · WL-8 execution integration · WL-9 admin console · WL-10 reporting · WL-11 deployment models.

Deprioritized relative to v0.1. The advisor-platform market is mature, consolidating, and defended by incumbents with entrenched contracts and migration costs. The realistic form of this option is an integration module inside an existing platform, not a competing platform. That is a partnership decision, not a build decision, and should not be specified further until a partner exists.

Entry preconditions: a named launch customer under contract, and a brokerage-as-a-service partner accepted through their diligence process.

7. Non-Functional Requirements (NFR)

Scoped by horizon — v0.1 applied enterprise NFRs from day one, which was premature.

ID	Requirement	Horizon
NFR-01	Publication job completes within 60 minutes of market close, with alert on failure or delay	0
NFR-02	Publication pipeline recovers automatically from transient failure; a missed day is alerted within 2 hours	0
NFR-03	Backups of the published record and PIT store, restore-tested quarterly	0
NFR-04	Encryption in transit (TLS 1.2+) and at rest for all subscriber data	1
NFR-05	No card data stored; PCI scope minimized via third-party processor	1
NFR-06	99.5% availability for subscriber-facing content	1
NFR-07	Privacy compliance (GDPR/CCPA) with subject access request handling	1
NFR-08	99.9% availability, externally measured	2
NFR-09	SOC 2 Type II	2
NFR-10	Annual third-party penetration test	2
NFR-11	Horizontal scalability to 50 tenants / 25,000 end users	2
NFR-12	WCAG 2.1 AA	2

8. Compliance Requirements (CMP)

ID	Requirement	Horizon
CMP-01	Counsel confirms impersonal, unpaid publication carries no registration requirement	0 — before first publication
CMP-02	Market data terms permit public display of derived rankings	0 — before TR-5
CMP-03	Counsel reviews the paid offering against the publisher's exclusion, including disclosure text	1 — before first paid subscriber
CMP-04	Performance presentation standard documented; live and hypothetical never blended	1
CMP-05	Market data redistribution agreements with per-recipient reporting	2 — before any external data delivery
CMP-06	Customer agreements define the vendor as technology/data provider with no discretionary authority	2
CMP-07	Model risk documentation for customer validation processes	2
CMP-08	Business continuity and vendor-risk package for due diligence questionnaires	2

9. Out of Scope

Permanently: custody of client assets; discretionary management; tax lot accounting and tax reporting.

Until Gate 1 passes: personalization of any kind; execution or order routing; API productization; multi-tenancy; SOC 2; retail direct-to-consumer marketing spend; non-US markets.

Anything in the second list appearing in a sprint before Gate 1 is scope creep, regardless of how small it looks.

10. Open Questions

ID	Question	Needed by
Q-01	Does unpaid impersonal publication require any registration in NJ or federally?	Before first publication
Q-02	Do current market data terms permit public display of derived rankings?	Before TR-5
Q-03	What is the minimum viable universe — full listed equities, or a constrained liquid subset?	Week 2
Q-04	Can 3 years of PIT history be reconstructed from available sources, or is a paid historical dataset required?	Week 4
Q-05	What is the acceptable weekly time commitment, and does the publication cadence fit inside it sustainably?	Before TR-1
Q-06	At Gate 1, is Horizon 1 as a standalone business preferable to entering Horizon 2 at all?	Month 18

11. Effort Envelope

Horizon	Build effort	External spend	Reversible?
0	Moderate — mostly refactoring existing code	Near zero; one counsel consultation	Fully
1	Moderate — subscription, delivery, disclosure	Low; counsel review, payment processing, hosting	Largely
2	Substantial	Material — data licences, SOC 2, counsel, possibly headcount	Not really

The asymmetry is the point. Horizon 0 costs little and can be abandoned at any time; Horizon 2 is a commitment. The gates exist so the commitment is only made against evidence.

12. Immediate Next Actions

1. Answer Q-01 and Q-02 — a single counsel consultation and a read of current data vendor terms.
2. Answer Q-03 and Q-05 — scope the universe to what can be published sustainably every trading day.
3. Build TR-1 and TR-2 to a minimum standard and **begin publishing**. Do not wait for TR-3 through TR-7 to be complete; a simpler record started sooner is worth more than a better one started later.
4. Backfill TR-3, TR-4, TR-6, TR-7 while the record accumulates.